

## BLAAVANDSHUK FYR, Denmark

WMO: 060810

Lat: 55.5575N

Lon: 8.0828E

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -7.2       | -5.2 | -10.1                         | 1.6 | -5.1 | -8.1 | 1.9 | -4.1 | 19.6                  | 4.8  | 17.9 | 4.4  | 4.9                   | 70   | 0.812 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 4.4                    | 24.6            | 18.6 | 22.3 | 17.9 | 20.7 | 17.2 | 19.7                | 22.8 | 18.7 | 21.5 | 17.8 | 20.1 | 5.9                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.5                            | 13.3 | 21.2 | 17.5 | 12.6 | 20.1 | 16.7 | 12.0 | 19.2 | 56.2          | 22.9 | 53.0 | 21.5 | 50.2 | 20.1 | 23.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.2          | 15.0 | 13.5 | DB                         | -9.2 | 27.6               | 3.1 | 2.0                                                | -11.5 | 29.1       | -13.3 | 30.3       | -15.1 | 31.4       | -17.4 | 32.8 | (4) |
| (5)               |      |      | WB                         | -9.5 | 21.2               | 3.1 | 1.2                                                | -11.8 | 22.1       | -13.6 | 22.8       | -15.3 | 23.5       | -17.6 | 24.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.0         | 2.0   | 1.7   | 3.4   | 6.9   | 10.7  | 13.9  | 16.5  | 16.9  | 14.4  | 10.5  | 6.5   | 3.7   | (6)  |  |
|                                                                        | DBStd       | 6.10        | 3.47  | 3.19  | 2.64  | 2.74  | 2.90  | 2.28  | 2.45  | 2.12  | 2.19  | 2.91  | 3.23  | 3.81  | (7)  |  |
|                                                                        | HDD10.0     | 1143        | 249   | 231   | 206   | 100   | 23    | 1     | 0     | 0     | 1     | 30    | 108   | 196   | (8)  |  |
|                                                                        | HDD18.3     | 3448        | 507   | 464   | 464   | 342   | 237   | 136   | 70    | 55    | 121   | 243   | 354   | 454   | (9)  |  |
|                                                                        | CDD10.0     | 765         | 0     | 0     | 0     | 8     | 46    | 116   | 201   | 214   | 131   | 45    | 4     | 0     | (10) |  |
|                                                                        | CDD18.3     | 28          | 0     | 0     | 0     | 0     | 1     | 2     | 13    | 11    | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 110         | 0     | 0     | 0     | 0     | 6     | 8     | 64    | 30    | 2     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 11          | 0     | 0     | 0     | 0     | 1     | 1     | 8     | 2     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.1         | 7.8   | 7.2   | 6.9   | 6.2   | 6.5   | 6.9   | 6.4   | 6.8   | 7.2   | 7.5   | 7.7   | 7.7   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 768         | 64    | 56    | 46    | 38    | 40    | 56    | 55    | 75    | 89    | 94    | 82    | 74    | (15) |  |
|                                                                        | PrecMax     | 975         | 125   | 101   | 96    | 71    | 72    | 105   | 125   | 137   | 183   | 195   | 146   | 142   | (16) |  |
|                                                                        | PrecMin     | 514         | 6     | 16    | 7     | 2     | 15    | 2     | 15    | 18    | 35    | 37    | 37    | 32    | (17) |  |
|                                                                        | PrecStd     | 116         | 32    | 25    | 25    | 17    | 18    | 33    | 32    | 33    | 46    | 36    | 33    | 32    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 8.7   | 7.3   | 12.3  | 18.5  | 24.0  | 24.2  | 27.8  | 26.4  | 22.7  | 16.6  | 13.5  | 10.4  | (19) |  |
|                                                                        |             | MCWB        | 7.7   | 6.0   | 8.4   | 12.3  | 16.9  | 16.9  | 19.7  | 20.0  | 17.8  | 15.0  | 12.6  | 9.3   | (20) |  |
|                                                                        | 2%          | DB          | 7.7   | 6.5   | 8.2   | 15.8  | 20.6  | 21.1  | 25.2  | 23.9  | 19.9  | 15.7  | 12.3  | 9.4   | (21) |  |
|                                                                        |             | MCWB        | 6.7   | 5.5   | 6.5   | 10.9  | 14.2  | 16.0  | 19.1  | 18.9  | 16.7  | 14.1  | 11.2  | 8.5   | (22) |  |
|                                                                        | 5%          | DB          | 6.8   | 5.9   | 7.1   | 13.3  | 17.7  | 19.0  | 22.5  | 21.8  | 18.5  | 15.0  | 11.4  | 8.7   | (23) |  |
|                                                                        |             | MCWB        | 5.8   | 5.1   | 6.0   | 9.7   | 13.3  | 15.7  | 18.4  | 18.4  | 15.9  | 13.5  | 10.1  | 7.8   | (24) |  |
|                                                                        | 10%         | DB          | 6.1   | 5.4   | 6.4   | 11.2  | 15.2  | 17.4  | 20.5  | 20.4  | 17.5  | 14.2  | 10.4  | 8.0   | (25) |  |
|                                                                        |             | MCWB        | 5.3   | 4.7   | 5.4   | 8.6   | 12.1  | 15.0  | 17.6  | 17.7  | 15.4  | 12.8  | 9.1   | 7.1   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 8.1   | 6.5   | 8.2   | 13.6  | 17.8  | 18.5  | 20.9  | 21.0  | 18.6  | 15.4  | 12.7  | 9.5   | (27) |  |
|                                                                        |             | MCDB        | 8.6   | 6.9   | 10.8  | 17.5  | 22.7  | 21.8  | 25.3  | 24.1  | 21.3  | 16.2  | 13.2  | 9.9   | (28) |  |
|                                                                        | 2%          | WB          | 6.9   | 5.8   | 7.0   | 11.4  | 15.1  | 17.1  | 19.7  | 19.9  | 17.2  | 14.3  | 11.5  | 8.8   | (29) |  |
|                                                                        |             | MCDB        | 7.4   | 6.3   | 7.8   | 14.5  | 18.7  | 19.4  | 23.2  | 22.5  | 19.0  | 15.0  | 12.1  | 9.2   | (30) |  |
|                                                                        | 5%          | WB          | 6.1   | 5.3   | 6.2   | 10.1  | 13.8  | 16.1  | 18.7  | 18.8  | 16.5  | 13.6  | 10.3  | 8.0   | (31) |  |
|                                                                        |             | MCDB        | 6.6   | 5.7   | 6.9   | 12.5  | 16.8  | 18.3  | 21.7  | 21.4  | 18.0  | 14.4  | 11.1  | 8.5   | (32) |  |
|                                                                        | 10%         | WB          | 5.4   | 4.9   | 5.6   | 9.0   | 12.5  | 15.2  | 17.8  | 18.0  | 15.7  | 13.0  | 9.4   | 7.3   | (33) |  |
|                                                                        |             | MCDB        | 6.0   | 5.3   | 6.2   | 10.5  | 14.7  | 17.2  | 20.2  | 20.0  | 17.2  | 13.8  | 10.2  | 7.9   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 3.0   | 2.9   | 3.7   | 5.1   | 5.1   | 4.4   | 4.6   | 4.4   | 4.3   | 3.7   | 3.2   | 3.2   | (35) |  |
|                                                                        |             | MCDBR       | 2.5   | 2.5   | 4.7   | 9.7   | 10.5  | 8.5   | 9.5   | 7.6   | 6.2   | 3.7   | 2.4   | 2.3   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 2.7   | 2.3   | 3.5   | 5.6   | 5.9   | 5.2   | 5.1   | 4.5   | 4.2   | 3.5   | 2.8   | 2.4   | (37) |  |
|                                                                        |             | MCWBR       | 2.4   | 2.1   | 4.0   | 7.8   | 8.6   | 6.8   | 7.7   | 6.4   | 5.0   | 3.4   | 2.4   | 2.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.327 | 0.344 | 0.364 | 0.397 | 0.387 | 0.385 | 0.397 | 0.400 | 0.380 | 0.359 | 0.336 | 0.313 | (40) |  |
|                                                                        |             | taud        | 2.247 | 2.305 | 2.312 | 2.270 | 2.369 | 2.397 | 2.379 | 2.377 | 2.412 | 2.384 | 2.300 | 2.245 | (41) |  |
|                                                                        | Ebn at Noon |             | 571   | 718   | 797   | 821   | 855   | 860   | 841   | 812   | 776   | 693   | 558   | 495   | (42) |  |
|                                                                        |             | Edh at Noon | 58    | 78    | 97    | 117   | 112   | 110   | 110   | 104   | 88    | 72    | 55    | 46    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.47  | 1.10  | 2.44  | 4.11  | 5.40  | 5.68  | 5.34  | 4.30  | 2.90  | 1.48  | 0.59  | 0.34  | (44) |  |
|                                                                        | RadStd      |             | 0.08  | 0.16  | 0.23  | 0.44  | 0.50  | 0.44  | 0.53  | 0.32  | 0.29  | 0.17  | 0.08  | 0.04  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A     | N/A     |         |
| (47) Regional (29 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | -110 | -152        | N/A     | N/A     |         |

Nomenclature: See separate page